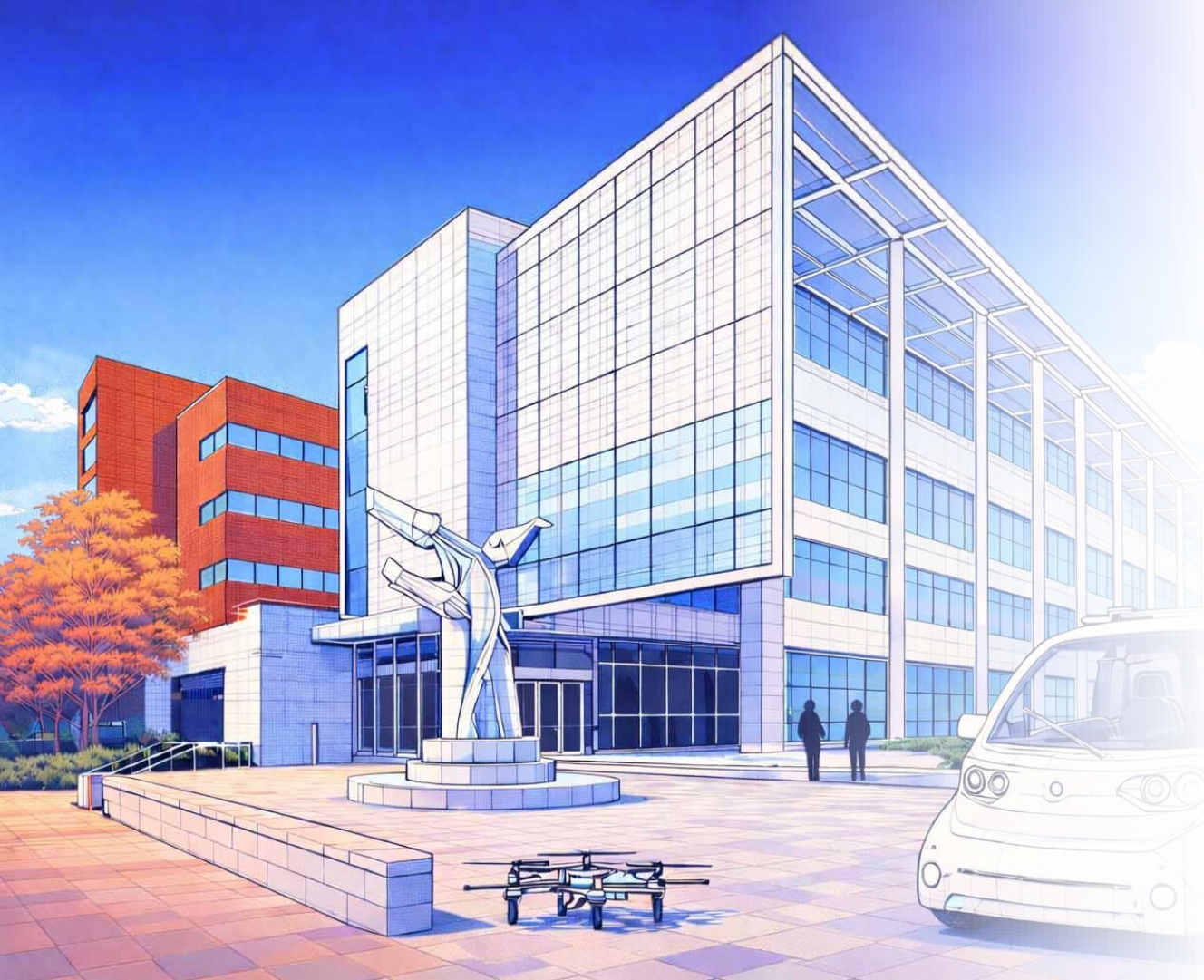




# ECE 484

## Lecture 6

### 3D Vision

Sayan Mitra

# Role of Perception in Autonomy

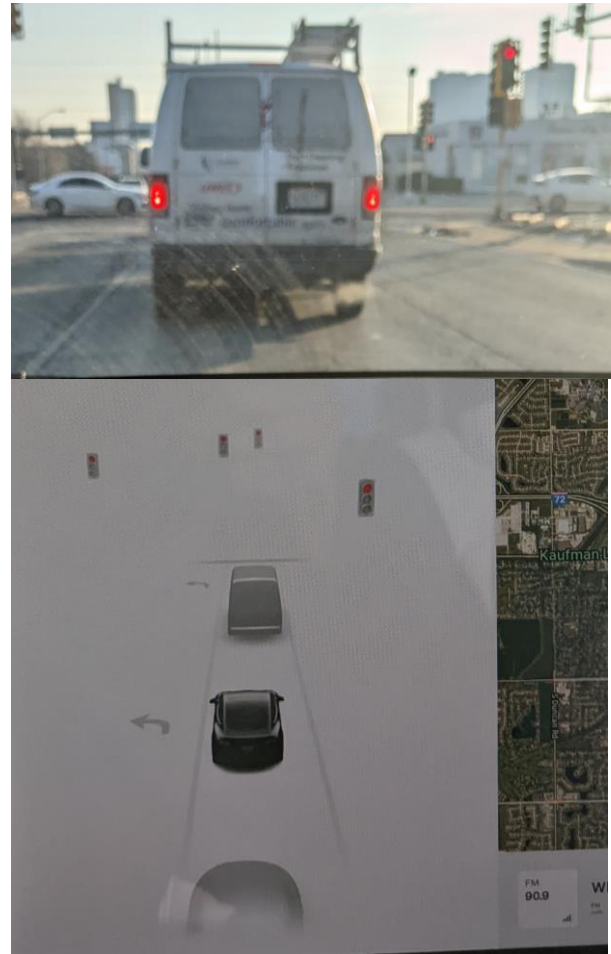
Perception module converts signals from the environment **state estimates** for the autonomous agent and its environment

Examples of state estimates:

- Type of lead vehicle, traffic sign
- **Position of ego on the map, relative to the lane, distance to the leading vehicle**
- Position of lead vehicle, speed, intention of the pedestrian

Types of estimates:

- Semantic: E.g., type of vehicle, sign
- Geometric: E.g., position, speed



# Problems

- Reconstructing the 3D structure of the scene from images
- Reconstructing 6D pose of camera from images
- Camera calibration

Input: image with points in pixels

Output:

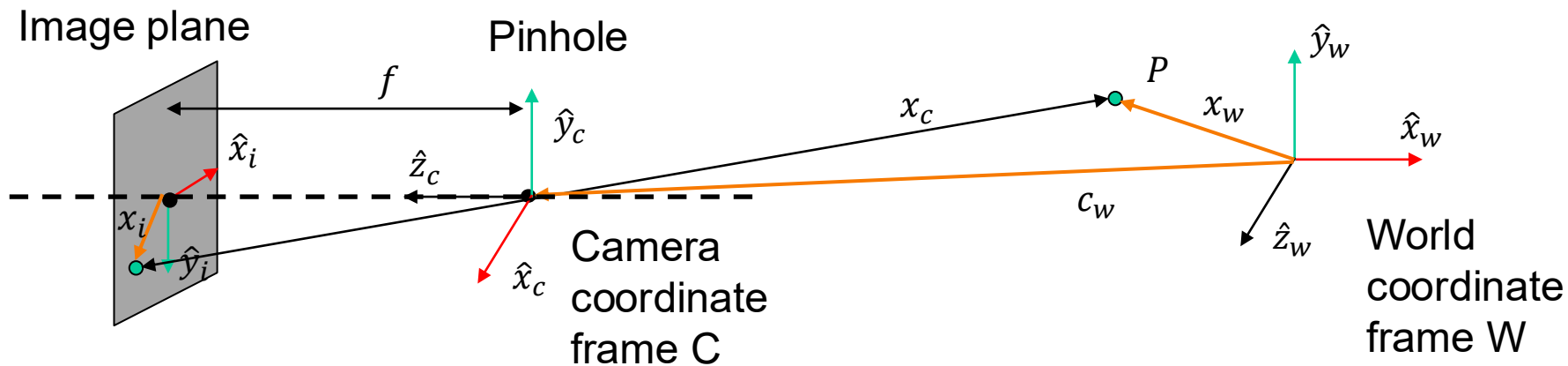
- Position of objects in millimeters in world frame
- Camera pose
- Camera intrinsic parameter

# Outline

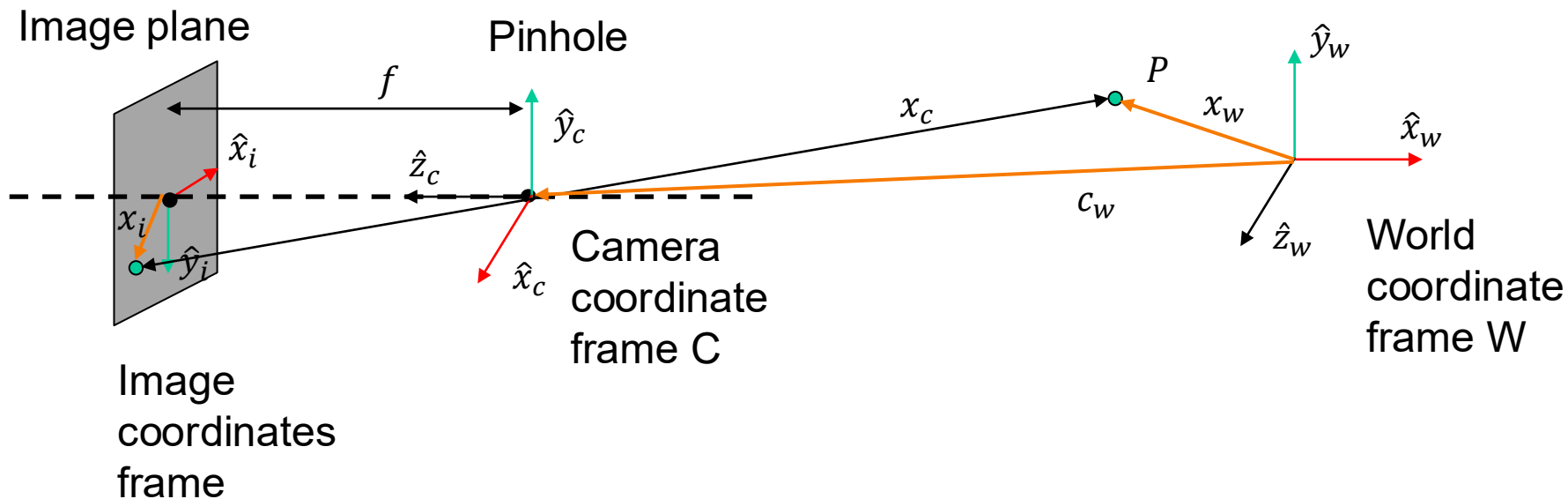
Forward camera model



# Forward Imaging Model: 3D to 2D

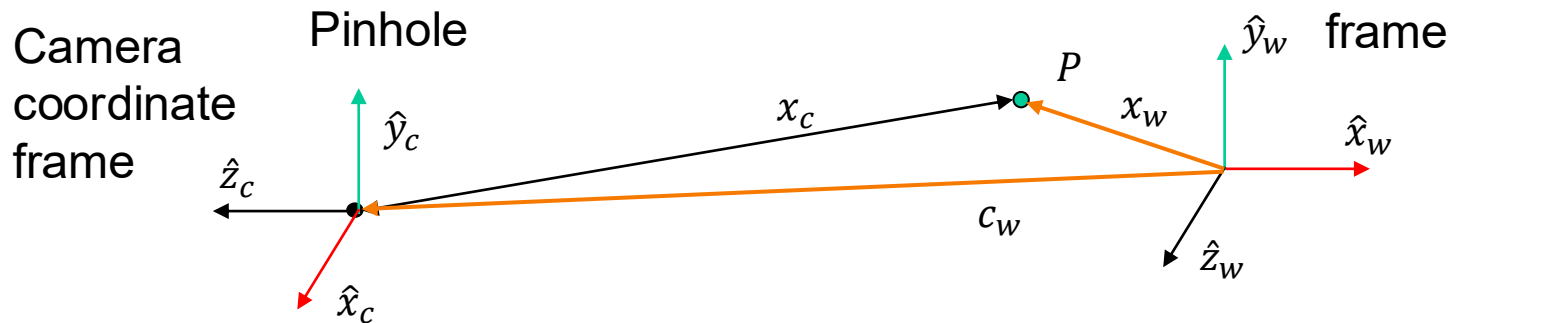


# Forward Imaging Model: 3D to 2D



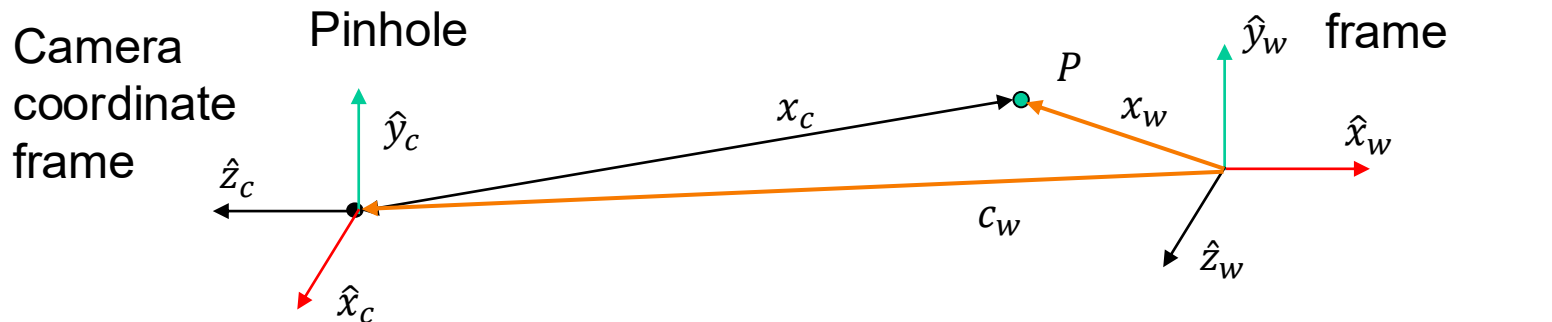
$$\mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \quad \text{3D-2D} \quad \mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad \text{3D-3D} \quad \mathbf{x}_w = \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix}$$

# World to camera Transformation (Extrinsic parameters)



Suppose we give you the position  $c_w$  of the camera's optical center and the its rotation  $R$  in the world coordinate frame (W) and the coordinates of  $P$  in the world coordinates  $P^w$  then what is the coordinate of  $P$  in the camera coordinates?

# World to camera Transformation (Extrinsic parameters)

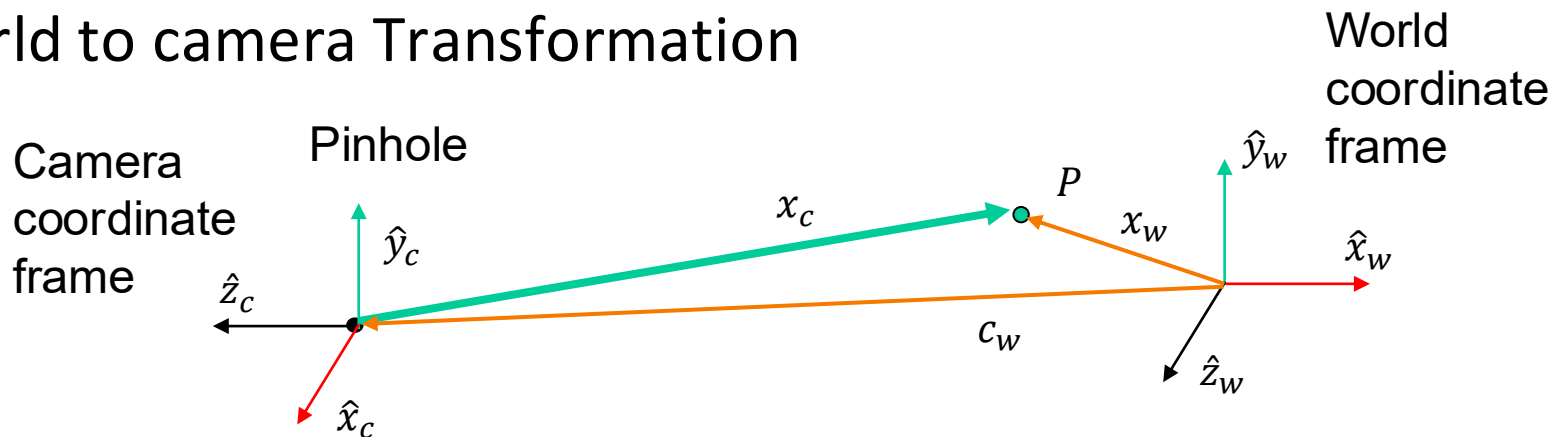


Position  $c_w$  and the orientation  $R$  of the camera rotation in the world coordinate frame (W) are the camera's **Extrinsic Parameters**

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \rightarrow \text{row 1 is the direction of } \hat{x}_c \text{ in world coordinates, 2 for } \hat{y}_c, \dots$$

This is a **rotation matrix**, i.e., the row vectors or the column vectors are orthonormal  
 $R^{-1} = R^T$  i.e.,  $R^T R = R R^T = I$

# World to camera Transformation



Position  $c_w$  and rotation  $R$  of the camera in the world coordinate frame (W)

These are camera's **Extrinsic Parameters**

**Recall**  $p^W = t_{WC} + R_{WC} p^C$   $p^C = R_{CW}(p^W - t_{WC})$

In the camera coordinate (c)  $x_c = R(x_w - c_w) = Rx_w - Rc_w = Rx_w + t$

define  $t = -Rc_w$

$$x_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \quad x_c = Rx_w + t$$

## Extrinsic Matrix

$$x_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

We have an affine transformation:  $x_c = Rx_w + t$

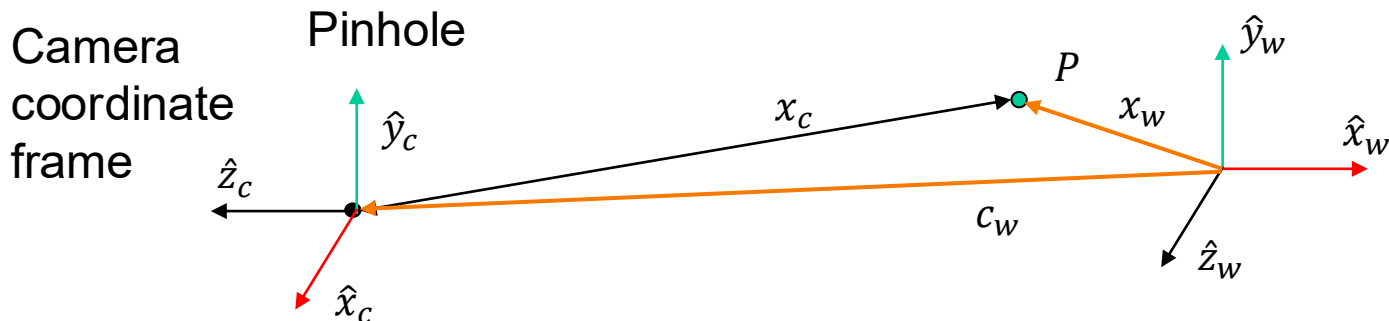
Can we represent it as  $x_c = Mx_w$ ? No

We can introduce a new coordinate  $\tilde{x}_c = [\tilde{x}, \tilde{y}, \tilde{z}, 1]^T$

Now can we represent this as a matrix multiplication  $\tilde{x}_c = M\tilde{x}_w$

$$\tilde{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

# Summary: Extrinsics define transformation from W to C frame



Given camera's extrinsic parameters  $(R, c_w)$ , the coordinates of P in camera coordinates  $x_c = R(x_w - c_w) = Rx_w - Rc_w = Rx_w + t$  with  $t = -Rc_w$

$$x_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \quad \text{Using homogeneous coordinates}$$

$$\tilde{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \quad \text{Extrinsic matrix } M_{ext} \quad \tilde{x}_c = M_{ext} \tilde{x}_w$$

# Geometry of Homogeneous coordinates (for 2D)

Affine transformation:  $x_c = Rx_w + t$

$$x_c = \begin{bmatrix} x_c \\ y_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

How to represent this as  $\tilde{x}_c = M\tilde{x}_w$

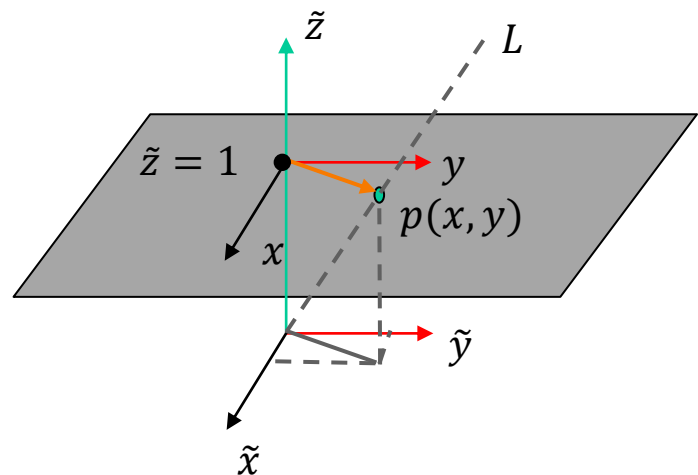
The homogeneous representation of a 2D point

$p = (x, y)$  is a 3D point  $\tilde{p} = (\tilde{x}, \tilde{y}, \tilde{z})$ .

The third coordinate  $\tilde{z} \neq 0$  is fictitious such that:

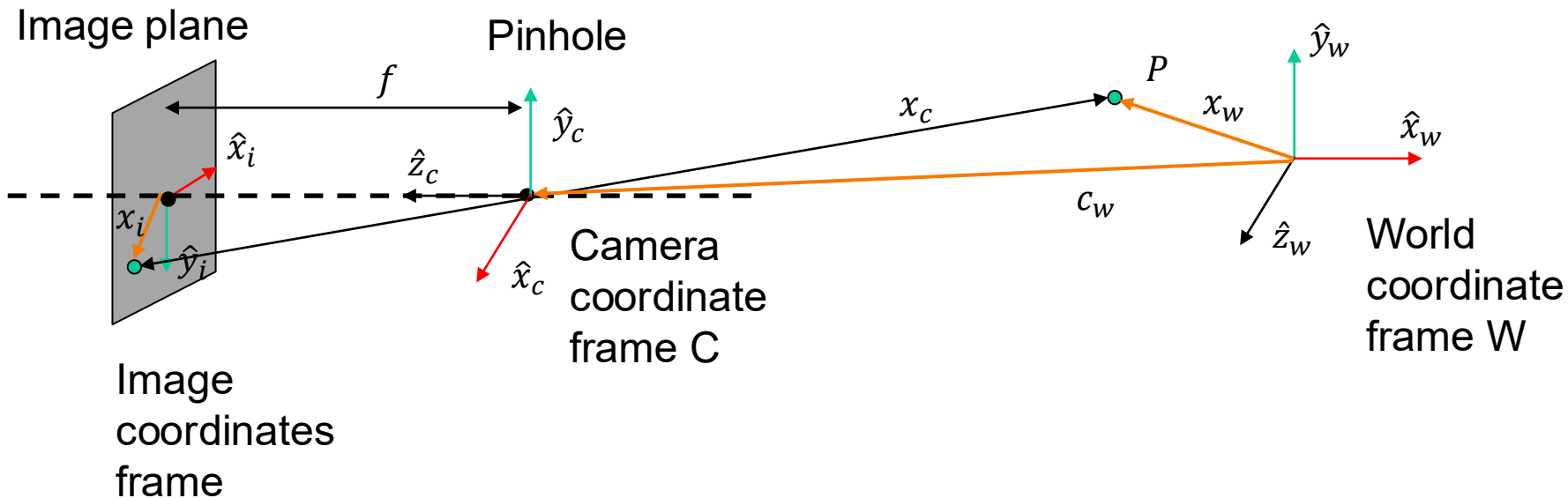
$$p = (x, y) \quad x = \frac{\tilde{x}}{\tilde{z}} \quad y = \frac{\tilde{y}}{\tilde{z}}$$
$$p \equiv \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{z}x \\ \tilde{z}y \\ \tilde{z} \end{bmatrix} \equiv \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = \tilde{p}$$

Geometric interpretation: all points on the line L  
(except origin) represent homogeneous  
coordinate  $p(x, y)$



$$p \equiv \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \equiv \begin{bmatrix} wx \\ \tilde{w}y \\ \tilde{w}z \\ \tilde{w} \end{bmatrix} \equiv \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \tilde{p}$$

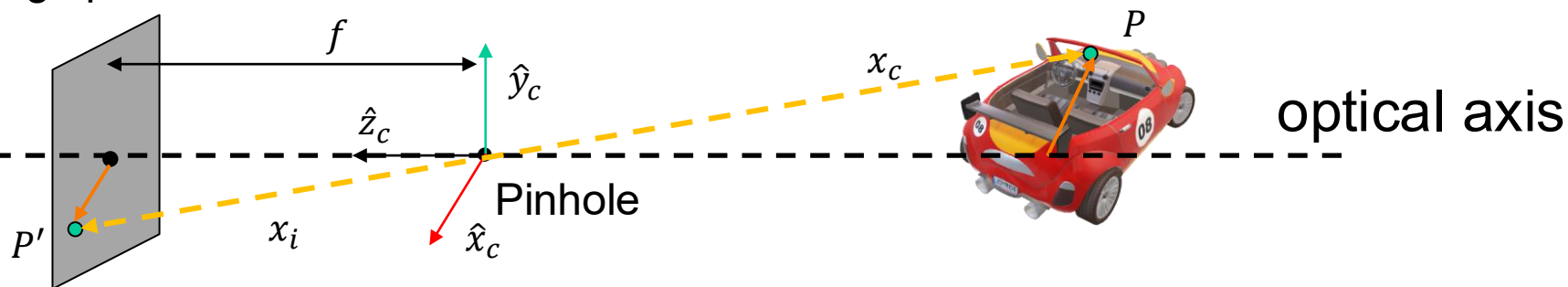
# Forward Imaging Model: 3D to 2D



$$\mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \quad \text{3D-2D} \quad \mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad \text{3D-3D} \quad \mathbf{x}_w = \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix}$$

# Perspective imaging with pinhole

Image plane

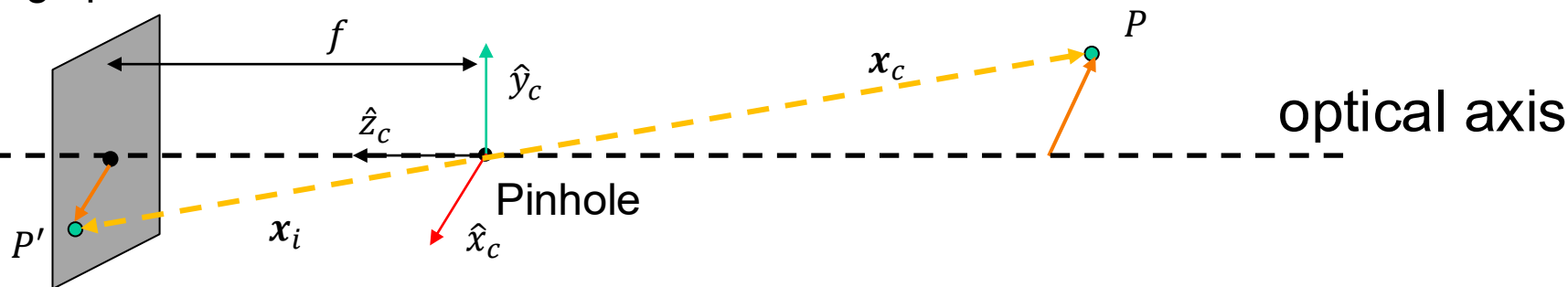


$f$ : Effective focal length

$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad \mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \\ f \end{bmatrix}$$

# Perspective imaging with pinhole

Image plane



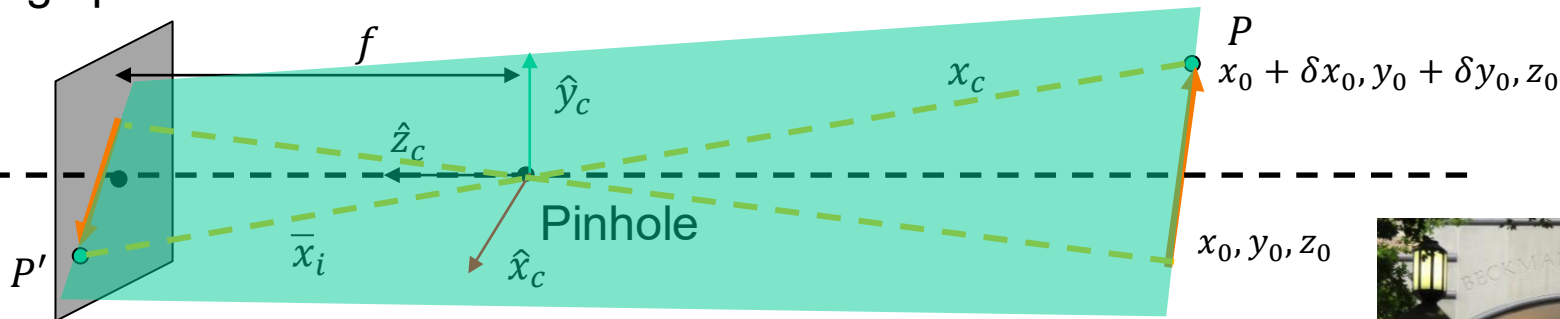
$f$ : Effective focal length

$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad \mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \\ f \end{bmatrix} \quad \boxed{\frac{\mathbf{x}_i}{f} = \frac{\mathbf{x}_c}{z_c}} \Rightarrow \boxed{\frac{x_i}{f} = \frac{x_c}{z_c}, \frac{y_i}{f} = \frac{y_c}{z_c}}$$

Not a linear transformation because division by  $z_c$

# Perspective projection of a line and magnification

Image plane



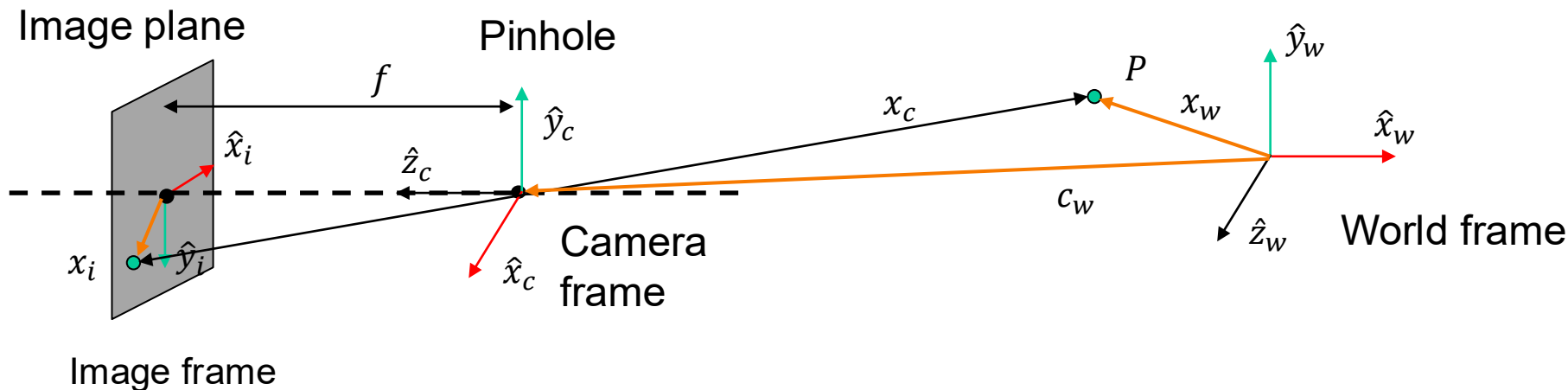
A line in 3D gets mapped to a line in the image plane

$$\frac{\bar{x}_i}{f} = \frac{x_c}{z_c} \Rightarrow \frac{x_i}{f} = \frac{x_c}{z_c}, \frac{y_i}{f} = \frac{y_c}{z_c}$$



Exercise: Show that magnification  $|m| = \frac{\text{object length}}{\text{image length}} = \frac{\sqrt{\delta x_i^2 + \delta y_i^2}}{\sqrt{\delta x_0^2 + \delta y_0^2}} = \left| \frac{f}{z_0} \right|$

# Camera coordinates to image plane coordinates



Perspective  
projection

$$\frac{x_i}{f} = \frac{x_c}{z_c} \text{ and } \frac{y_i}{f} = \frac{y_c}{z_c}$$

$$x_i = f \frac{x_c}{z_c} \text{ and } y_i = f \frac{y_c}{z_c}$$

# Image plane to image sensor mapping

Image plane

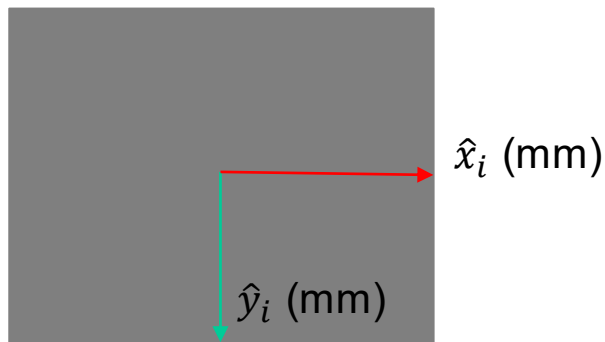
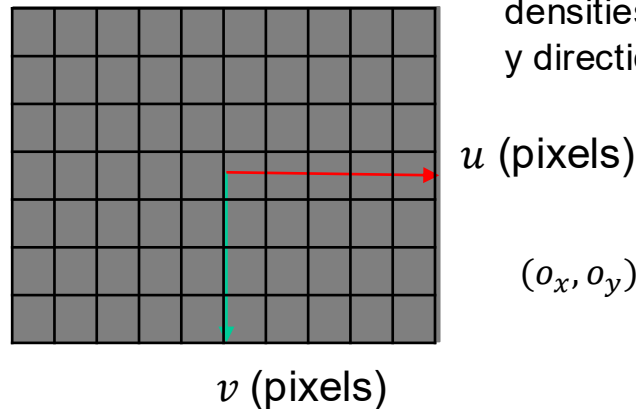


Image sensor



Pixels may be rectangular  
Let  $m_x$  and  $m_y$  be the pixel  
densities (pixels/mm) in x and  
y directions

$(o_x, o_y)$  Principle point

$$x_i = f \frac{x_c}{z_c} \text{ and } y_i = f \frac{y_c}{z_c}$$

$$u = m_x f \frac{x_c}{z_c} \text{ and } v = m_y f \frac{y_c}{z_c}$$

$$u = m_x f \frac{x_c}{z_c} + o_x \text{ and } v = m_y f \frac{y_c}{z_c} + o_y$$

$$u = f_x \frac{x_c}{z_c} + o_x \text{ and } v = f_y \frac{y_c}{z_c} + o_y$$

Intrinsic parameters:  $f_x, f_y, o_x, o_y$

# Nonlinear to linear model using homogeneous coordinates

$$u = f_x \frac{x_c}{z_c} + o_x \text{ and } v = f_y \frac{y_c}{z_c} + o_y$$

$$uz_c = f_x x_c + o_x z_c \text{ and } vz_c = f_y y_c + o_y z_c$$

$$\text{Adding } z_c = 0 \cdot x_c + 0 \cdot y_c + 1 \cdot z_c$$

Stacking them

$$\mathbf{u} \equiv \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \equiv \begin{bmatrix} z_c u \\ z_c v \\ z_c \end{bmatrix} = \begin{bmatrix} f_x x_c + z_c o_x \\ f_y y_c + z_c o_y \\ z_c \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

Homogeneous representation of  $(u, v)$  as a 3D point  $\tilde{u} = (\tilde{u}, \tilde{v}, \tilde{w})$

$$(uz_c, vz_c, z_c) \equiv (u, v, 1)$$

Linear model of perspective projection  $\tilde{u} = [K|0]\tilde{x}_c = M_{int}\tilde{x}_c$

**Intrinsic matrix** ( $M_{int}$ )

**Calibration matrix**  $K$  (upper right triangular)

# Forward Camera Model

Camera to pixel

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

World to camera

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\tilde{u} = M_{int} \tilde{x}_w = [K|0] \tilde{x}_w$$

$$\tilde{x}_c = M_{ext} \tilde{x}_w$$

$$\tilde{u} = M_{int} M_{ext} \tilde{x}_w = P \tilde{x}_w = K[R|t] \tilde{x}_w$$

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

P: **Projection matrix**

The left 3x3 matrix in P is a product of an upper triangular and an orthonormal matrix



# Outline

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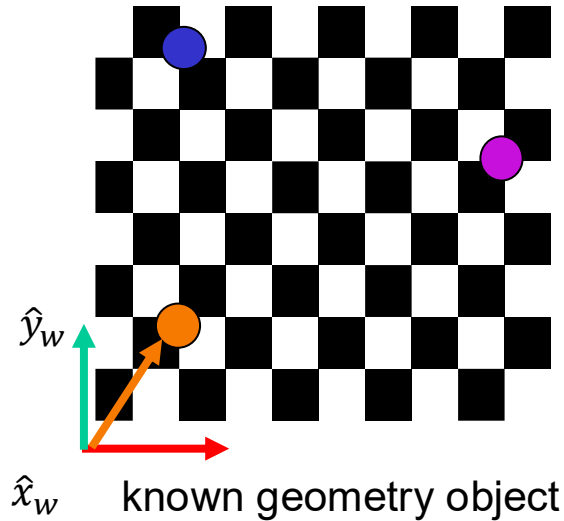
Forward camera model

Calibration

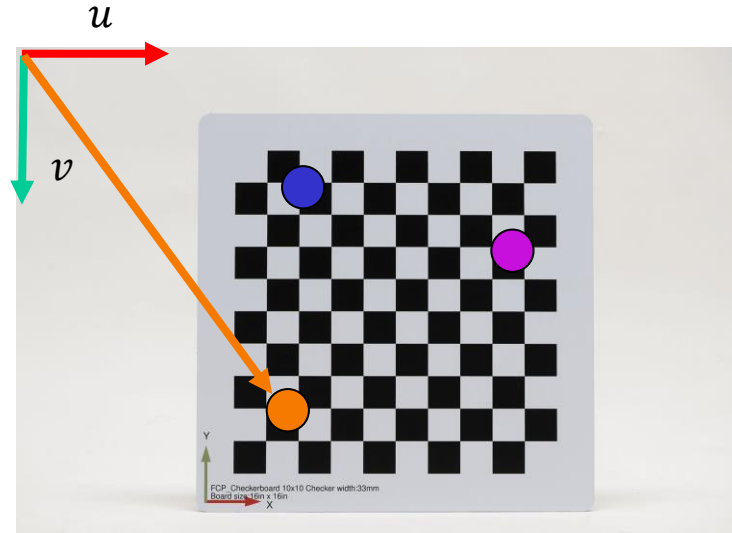


# Camera Calibration Procedure

Step 1. Capture image of object with known geometry



$$\bullet \mathbf{x}_w = \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix}$$



captured image

$$\bullet \mathbf{u} = \begin{bmatrix} u \\ v \end{bmatrix}$$

# Camera Calibration

Step 2. For each point  $i$  with  $(x_w^{(i)}, y_w^{(i)}, z_w^{(i)})$  we get a linear equation

$$\begin{bmatrix} u^{(i)} \\ v^{(i)} \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w^{(i)} \\ y_w^{(i)} \\ z_w^{(i)} \\ 1 \end{bmatrix}$$

Step 3. Collecting many  $u^{(i)} = \frac{p_{11}x_w^{(i)} + p_{12}y_w^{(i)} + p_{13}z_w^{(i)} + p_{14}}{p_{31}x_w^{(i)} + p_{32}y_w^{(i)} + p_{33}z_w^{(i)} + p_{34}}$  points and

rearranging  $p = [p_{11} \ p_{12} \ \dots \ p_{34}]^T$  as a vector  $\mathbb{R}^{12}$  we get  $A\mathbf{p} = 0$  where a single row of  $A_{u,i} = [x_w^{(i)} \ y_w^{(i)} \ z_w^{(i)} \ 1 \ 0 \ 0 \ 0 \ -v_i x_w^{(i)} \ -v_i y_w^{(i)} \ -v_i z_w^{(i)} \ -v_i]$

Step 4. Solve for  $\mathbf{p}$

$$A\mathbf{p} = \mathbf{0}$$

Find  $\mathbf{p}$  such that  $A\mathbf{p} = \mathbf{0}$

Find a vector in the null space of  $A$

With noisy measurements the null space will collapse to  $\mathbf{p} = \mathbf{0}$

Also, remember we can only find  $\mathbf{p}$  up to scale

- We can set one of the elements to be 1 arbitrarily OR
- Add  $\|\mathbf{p}\| = 1$  as a constraint

# Projection matrix scale

Since projection matrix works on homogeneous coordinates

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \equiv k \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix}$$

Therefore

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} = k \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

Therefore, Projection Matrices P and kP produce the same homogenous pixel coordinates

Projection matrix is defined only upto a scale factor

Scaling the world and the camera will produce indistinguishable images

That is , we can only find the projection matrix up to scale; we choose  $\|p\| = 1$

# Least Squares Solution for Projection Matrix

We want  $A\mathbf{p}$  as close to 0 as possible and  $\|\mathbf{p}\|^2 = 1$

$$\min_{\mathbf{p}} \|A\mathbf{p}\|^2 \text{ such that } \|\mathbf{p}\|^2 = 1$$

$$\min_{\mathbf{p}} \left| \mathbf{p}^T A^T A \mathbf{p} \right|^2 \text{ such that } \mathbf{p}^T \mathbf{p} = 1$$

$$L(\mathbf{p}, \lambda) = \mathbf{p}^T A^T A \mathbf{p} - \lambda(\mathbf{p}^T \mathbf{p} - 1). \text{ Unconstrained optimization}$$

Taking derivative  $\frac{\partial L}{\partial \mathbf{p}} = 0$  gives  $2A^T A \mathbf{p} - 2\lambda \mathbf{p} = \mathbf{0}$

$$A^T A \mathbf{p} = \lambda \mathbf{p}$$

$\mathbf{p}$  is the Eigenvector corresponding to the smallest eigenvalue of  $A^T A$

Rearrange  $\mathbf{p}$  to get the projection matrix  $\mathbf{P}$

## From projection matrix to Mint Mext

1. Reshape  $P \in \mathbb{R}^{3 \times 4}$   $P = [M \mid \mathbf{p}_4]$  with  $M \in \mathbb{R}^{3 \times 3}$
2. Factor M into intrinsic  $K$  and rotation  $R$  using RQ factorization:  
 $M = K R$  where  $K$  is upper triangular  $R$  is orthonormal
3. Solve for translation

From  $P = K [R \mid t] = [K R \mid K t]$  gives:

$$\mathbf{p}_4 = K t \Rightarrow t = K^{-1} \mathbf{p}_4$$

From data we have derived the intrinsic  $M_{int} = [K|0]$  and the extrinsic  $M_{ext} = [R|t]$  matrices and we have a fully calibrated camera

# 3x3 Affine Transformations

Scaling

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

Skew

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} 1 & m_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

Translation

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

Rotation

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

Composition of Transformations

General form

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

# Projective Transformations

General form 
$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix} \tilde{p}_2 = H \tilde{p}_1$$

$H$  is called a homography

Origin does not necessarily map to origin

Lines map to lines

Parallel lines do not necessarily remain parallel

Closed under composition

Homographies are defined up to scaling

